Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: Germano Soares dba Sandy Mush Crosses

Physical address of dairy:

Number and Street	City	County	Zip Code
5241 W Sandy Mush RD	Merced	Merced	95340

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 07/01/1963

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

X074-X080-X025-XXXX

B. OPERATORS

Soares, Germano L

Operator name: Soares, Germano L

Telephone no.: Landline: ; Cellular: (209) 000-0000

Mailing Address Number and Street	City	State	Zip Code
5241 W Sandy Mush RD	Merced	CA	95341

This operator is responsible for paying permit fees.

C. OWNERS

Soares, Germano L

Legal owner name: Soares, Germano L

Telephone no.: Landline: ; Cellular: (209) 000-0000

Mailing Address Number and Street	City	State	Zip Code
5241 W Sandy Mush RD	Merced	CA	95341

This owner is responsible for paying permit fees.

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AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	85	70	80	70	100
Number under roof	750	0	0	0	0	0
Maximum number	750	85	70	80	70	100
Average number	750	85	70	80	70	100
Avg live weight (lbs)	1,000	1,050	700	600		

Predominant milk cow breed: Jersey

Average milk production: 60 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 21,441.83 *tons per reporting period*

Total nitrogen from manure: 278,461.12 *lbs per reporting period*

After ammonia losses (30% loss applied): 194,922.78 *lbs per reporting period*

Total phosphorus from manure: 45,538.23 *lbs per reporting period*

Total potassium from manure: 138,704.42 *lbs per reporting period*

Total salt from manure: 372,683.25 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 7,032,000 *gallons*

Total nitrogen generated: 25,209.94 *lbs*

Total phosphorus generated: 3,168.61 *lbs*

Total potassium generated: 41,030.31 *lbs*

Total salt generated: 171,139.26 *lbs*

	7,032,000	*gallons applied*
+	0	*gallons exported*
-	0	*gallons imported*
=	7,032,000	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
Ag Well	Ground water
Domestic Well/ barn	Ground water
Domestic Well/ house	Ground water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

No dry manure nutrient imports entered.

No process wastewater nutrient imports entered.

No commercial or other nutrient imports entered.

G. NUTRIENT EXPORTS

No solid nutrient exports entered.

No liquid nutrient exports entered.

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
#1	35	35	2	process wastewater	X074-X080-X025-XXXX
#2	33	33	2	process wastewater	X074-X080-X025-XXXX
#3	40	40	2	process wastewater	X074-X080-X025-XXXX
#4	10	10	2	process wastewater	X074-X080-X025-XXXX
Totals for areas that were used for application	118	118	8		
Totals for areas that were not used for application					
Land application area totals	118	118	8		

B. CROPS AND HARVESTS

#1

Field name: #1

11/05/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 35 Plant date: 11/05/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/28/2023	630.00 *ton*	Dry-weight		54.7	11,100.00	1,600.00	10,500.00		10.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	18.00	181.02	26.09	171.23	1,679.72

06/15/2023: Corn, silage

Crop: Corn, silage Acres planted: 35 Plant date: 06/15/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/01/2023	875.00 *ton*	Dry-weight		0.1	0.00	0.00	0.00		0.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	25.00	0.00	0.00	0.00	0.00

#2

Field name: #2

11/05/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 33 Plant date: 11/05/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/28/2023	594.00 *ton*	Dry-weight		66.1	18,500.00	2,150.00	15,000.00		14.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	18.00	225.77	26.24	183.06	1,818.40

06/15/2023: Corn, silage

Crop: Corn, silage Acres planted: 33 Plant date: 06/15/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/01/2023	825.00 *ton*	Dry-weight		0.1	0.00	0.00	0.00		0.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	25.00	0.00	0.00	0.00	0.00

#3

Field name: #3

11/05/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 40 Plant date: 11/05/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/29/2023	720.00 *ton*	Dry-weight		63.9	15,200.00	2,050.00	14,700.00		12.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	18.00	197.54	26.64	191.04	1,676.48

#3

06/15/2023: Corn, silage

Crop: Corn, silage Acres planted: 40 Plant date: 06/15/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/01/2023	1,000.00 *ton*	Dry-weight		0.1	0.00	0.00	0.00		0.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	25.00	0.00	0.00	0.00	0.00

#4

Field name: #4

11/05/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 10 Plant date: 11/05/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/29/2023	180.00 *ton*	Dry-weight		0.1	0.00	0.00	0.00		0.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	10.00	100.00	16.00	83.00	0.00
Total actual harvest content	18.00	0.00	0.00	0.00	0.00

06/15/2023: Corn, silage

Crop: Corn, silage Acres planted: 10 Plant date: 06/15/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/01/2023	250.00 *ton*	Dry-weight		0.1	0.00	0.00	0.00		0.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	25.00	0.00	0.00	0.00	0.00

NUTRIENT BUDGET

A. LAND APPLICATIONS

#1 - 11/05/2022: Oats, silage-soft dough

Field name: #1

Crop: Oats, silage-soft dough

Plant date: 11/05/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	56.65	6.06	73.65	412.00	432,000.00 *gal*
Ag Well	Ground water	8.49	0.00	0.00	510.89	3,456,000.00 *gal*
Application event totals		65.14	6.06	73.65	922.89	

#1 - 06/15/2023: Corn, silage

Field name: #1

Crop: Corn, silage

Plant date: 06/15/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	22.93	2.06	48.75	192.27	288,000.00 *gal*
Ag Well	Ground water	12.73	0.00	0.00	766.33	5,184,000.00 *gal*
Application event totals		35.67	2.06	48.75	958.60	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well	Ground water	14.15	0.00	0.00	851.48	5,760,000.00 *gal*
Application event totals		14.15	0.00	0.00	851.48	

#1 - 06/15/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	17.20	1.55	36.57	144.20	216,000.00 *gal*
Ag Well	Ground water	9.55	0.00	0.00	574.75	3,888,000.00 *gal*
Application event totals		26.75	1.55	36.57	718.95	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	17.20	1.55	36.57	144.20	216,000.00 *gal*
Ag Well	Ground water	9.55	0.00	0.00	574.75	3,888,000.00 *gal*
Application event totals		26.75	1.55	36.57	718.95	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	19.57	3.24	29.14	103.00	180,000.00 *gal*
Ag Well	Ground water	7.96	0.00	0.00	478.96	3,240,000.00 *gal*
Application event totals		27.53	3.24	29.14	581.96	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well	Ground water	7.07	0.00	0.00	425.74	2,880,000.00 *gal*
Application event totals		7.07	0.00	0.00	425.74	

#2 - 11/05/2022: Oats, silage-soft dough

Field name: #2

Crop: Oats, silage-soft dough

Plant date: 11/05/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

#2 - 11/05/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	60.08	6.42	78.11	436.97	432,000.00 *gal*
Ag Well	Ground water	9.00	0.00	0.00	541.85	3,456,000.00 *gal*
Application event totals		69.09	6.42	78.11	978.82	

#2 - 06/15/2023: Corn, silage

Field name: #2

Crop: Corn, silage Plant date: 06/15/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	24.32	2.18	51.71	203.92	288,000.00 *gal*
Ag Well	Ground water	13.50	0.00	0.00	812.77	5,184,000.00 *gal*
Application event totals		37.83	2.18	51.71	1,016.69	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	24.32	2.18	51.71	203.92	288,000.00 *gal*
Ag Well	Ground water	14.69	0.00	0.00	884.27	5,640,000.00 *gal*
Application event totals		39.02	2.18	51.71	1,088.19	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	12.67	1.14	26.93	106.21	150,000.00 *gal*
Ag Well	Ground water	9.69	0.00	0.00	583.24	3,720,000.00 *gal*
Application event totals		22.36	1.14	26.93	689.45	

#2 - 06/15/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	17.30	2.86	25.76	91.04	150,000.00 *gal*
Ag Well	Ground water	9.69	0.00	0.00	583.24	3,720,000.00 *gal*
Application event totals		26.99	2.86	25.76	674.28	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	13.84	2.29	20.60	72.83	120,000.00 *gal*
Ag Well	Ground water	8.13	0.00	0.00	489.17	3,120,000.00 *gal*
Application event totals		21.96	2.29	20.60	562.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well	Ground water	5.63	0.00	0.00	338.66	2,160,000.00 *gal*
Application event totals		5.63	0.00	0.00	338.66	

#3 - 11/05/2022: Oats, silage-soft dough

Field name: #3

Crop: Oats, silage-soft dough

Plant date: 11/05/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	66.09	7.07	85.92	480.67	576,000.00 *gal*
Ag Well	Ground water	9.80	0.00	0.00	589.82	4,560,000.00 *gal*
Application event totals		75.89	7.07	85.92	1,070.50	

#3 - 06/15/2023: Corn, silage

#3 - 06/15/2023: Corn, silage

Field name: #3

Crop: Corn, silage

Plant date: 06/15/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	25.09	2.25	53.32	210.29	360,000.00 *gal*
Ag Well	Ground water	13.92	0.00	0.00	838.17	6,480,000.00 *gal*
Application event totals		39.01	2.25	53.32	1,048.47	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well	Ground water	15.47	0.00	0.00	931.30	7,200,000.00 *gal*
Application event totals		15.47	0.00	0.00	931.30	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	20.07	1.80	42.66	168.24	288,000.00 *gal*
Ag Well	Ground water	11.14	0.00	0.00	670.54	5,184,000.00 *gal*
Application event totals		31.21	1.80	42.66	838.77	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	20.07	1.80	42.66	168.24	288,000.00 *gal*
Ag Well	Ground water	1.24	0.00	0.00	74.50	576,000.00 *gal*
Application event totals		21.31	1.80	42.66	242.74	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well	Ground water	8.35	0.00	0.00	502.90	3,888,000.00 *gal*
Application event totals		8.35	0.00	0.00	502.90	

#3 - 06/15/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	205.49	34.02	305.98	1,081.51	2,160,000.00 *gal*
Ag Well	Ground water	8.77	0.00	0.00	527.74	4,080,000.00 *gal*
Application event totals		214.25	34.02	305.98	1,609.25	

#4 - 11/05/2022: Oats, silage-soft dough

Field name: #4

Crop: Oats, silage-soft dough Plant date: 11/05/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	66.09	7.07	85.92	480.67	144,000.00 *gal*
Ag Well	Ground water	9.80	0.00	0.00	589.82	1,140,000.00 *gal*
Application event totals		75.89	7.07	85.92	1,070.50	

#4 - 06/15/2023: Corn, silage

Field name: #4

Crop: Corn, silage Plant date: 06/15/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	40.14	3.61	85.32	336.47	144,000.00 *gal*
Ag Well	Ground water	22.28	0.00	0.00	1,341.07	2,592,000.00 *gal*
Application event totals		62.42	3.61	85.32	1,677.55	

#4 - 06/15/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well	Ground water	24.75	0.00	0.00	1,490.08	2,880,000.00 *gal*
Application event totals		24.75	0.00	0.00	1,490.08	

07/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	30.10	2.70	63.99	252.35	108,000.00 *gal*
Ag Well	Ground water	16.71	0.00	0.00	1,005.81	1,944,000.00 *gal*
Application event totals		46.81	2.70	63.99	1,258.16	

07/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	30.10	2.70	63.99	252.35	108,000.00 *gal*
Ag Well	Ground water	16.71	0.00	0.00	1,005.81	1,944,000.00 *gal*
Application event totals		46.81	2.70	63.99	1,258.16	

08/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	36.53	6.05	54.40	192.27	96,000.00 *gal*
Ag Well	Ground water	14.85	0.00	0.00	894.05	1,728,000.00 *gal*
Application event totals		51.38	6.05	54.40	1,086.32	

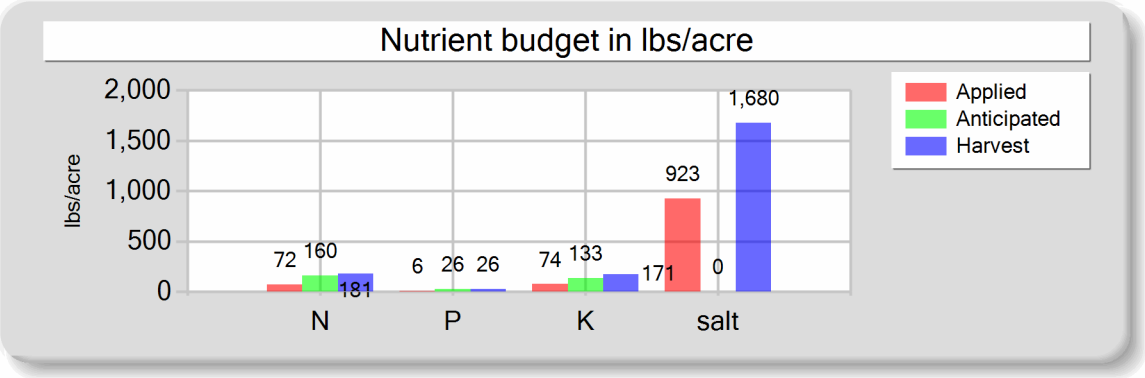
08/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well	Ground water	12.38	0.00	0.00	745.04	1,440,000.00 *gal*
Application event totals		12.38	0.00	0.00	745.04	

B. NUTRIENT BUDGET

#1 - 11/05/2022: Oats, silage-soft dough

Field name: #1 Crop: Oats, silage-soft dough Plant date: 11/05/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	56.65	6.06	73.65	412.00
Fresh water	8.49	0.00	0.00	510.89
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	72.14	6.06	73.65	922.89
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	181.02	26.09	171.23	1,679.72
Nutrient balance	-108.88	-20.04	-97.59	-756.83
Applied to removed ratio	0.40	0.23	0.43	0.55

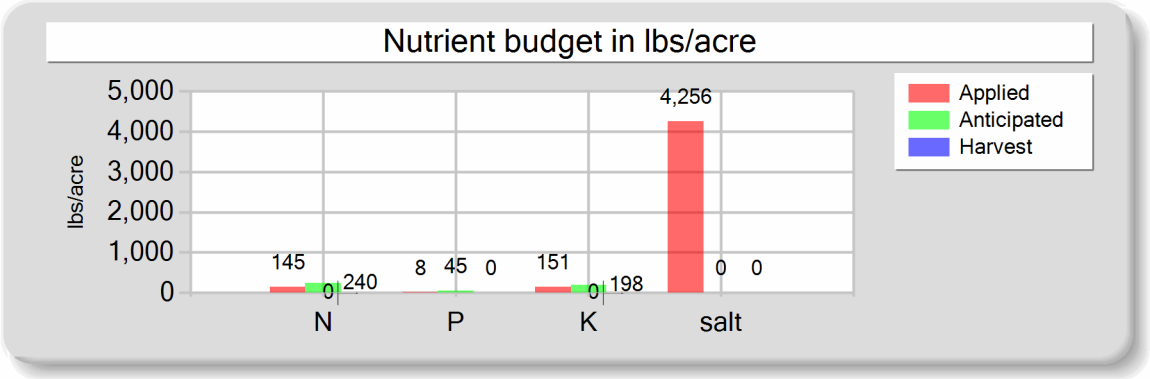
Fresh water applied	
3,456,000.00	*gallons*
127.27	*acre-inches*
3.64	*inches/acre*

Process wastewater applied	
432,000.00	*gallons*
15.91	*acre-inches*
0.45	*inches/acre*

Total harvests for the crop	
1	*harvests*

#1 - 06/15/2023: Corn, silage

Field name: #1 Crop: Corn, silage Plant date: 06/15/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	76.91	8.39	151.03	583.67
Fresh water	61.00	0.00	0.00	3,671.99
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	144.91	8.39	151.03	4,255.66
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	0.00	0.00	0.00	0.00
Nutrient balance	144.91	8.39	151.03	4,255.66
Applied to removed ratio	0.00	0.00	0.00	0.00

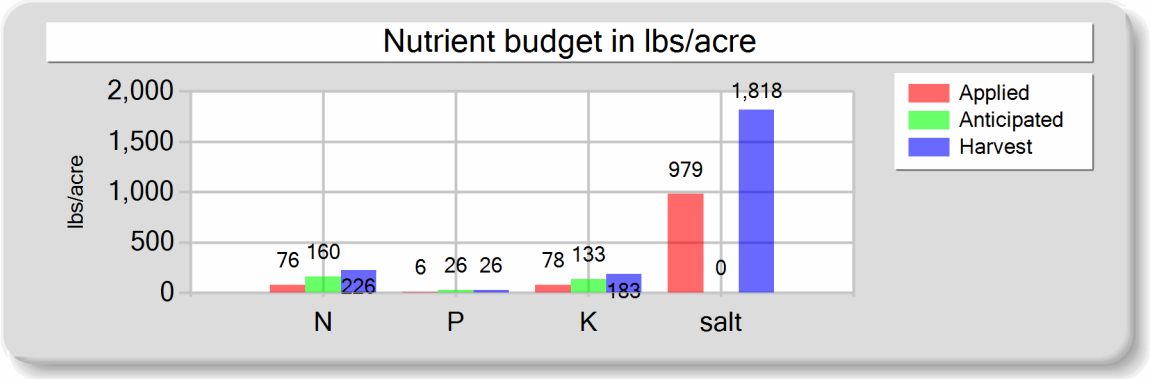
Fresh water applied	
24,840,000.00	*gallons*
914.77	*acre-inches*
26.14	*inches/acre*

Process wastewater applied	
900,000.00	*gallons*
33.14	*acre-inches*
0.95	*inches/acre*

Total harvests for the crop	
1	*harvests*

#2 - 11/05/2022: Oats, silage-soft dough

Field name: #2 Crop: Oats, silage-soft dough Plant date: 11/05/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	60.08	6.42	78.11	436.97
Fresh water	9.00	0.00	0.00	541.85
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	76.09	6.42	78.11	978.82
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	225.77	26.24	183.06	1,818.40
Nutrient balance	-149.69	-19.82	-104.95	-839.57
Applied to removed ratio	0.34	0.24	0.43	0.54

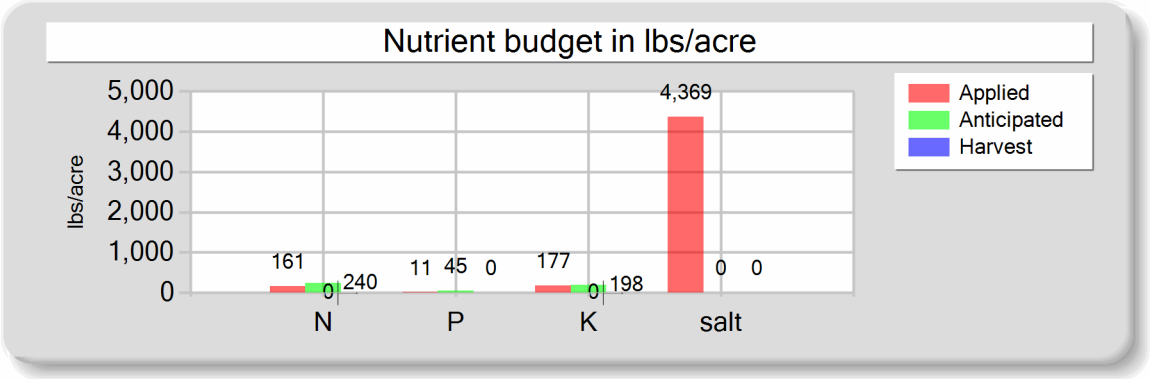
Fresh water applied	
3,456,000.00	*gallons*
127.27	*acre-inches*
3.86	*inches/acre*

Process wastewater applied	
432,000.00	*gallons*
15.91	*acre-inches*
0.48	*inches/acre*

Total harvests for the crop	
1	*harvests*

#2 - 06/15/2023: Corn, silage

Field name: #2 Crop: Corn, silage Plant date: 06/15/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	92.45	10.66	176.71	677.92
Fresh water	61.32	0.00	0.00	3,691.34
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	160.78	10.66	176.71	4,369.26
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	0.00	0.00	0.00	0.00
Nutrient balance	160.78	10.66	176.71	4,369.26
Applied to removed ratio	0.00	0.00	0.00	0.00

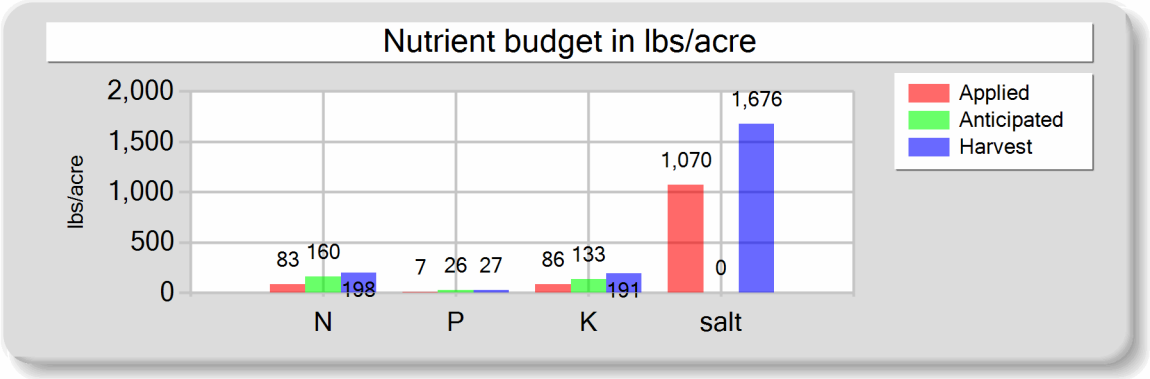
Fresh water applied	
23,544,000.00	*gallons*
867.05	*acre-inches*
26.27	*inches/acre*

Process wastewater applied	
996,000.00	*gallons*
36.68	*acre-inches*
1.11	*inches/acre*

Total harvests for the crop	
1	*harvests*

#3 - 11/05/2022: Oats, silage-soft dough

Field name: #3 Crop: Oats, silage-soft dough Plant date: 11/05/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	66.09	7.07	85.92	480.67
Fresh water	9.80	0.00	0.00	589.82
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	82.89	7.07	85.92	1,070.50
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	197.54	26.64	191.04	1,676.48
Nutrient balance	-114.65	-19.58	-105.12	-605.99
Applied to removed ratio	0.42	0.27	0.45	0.64

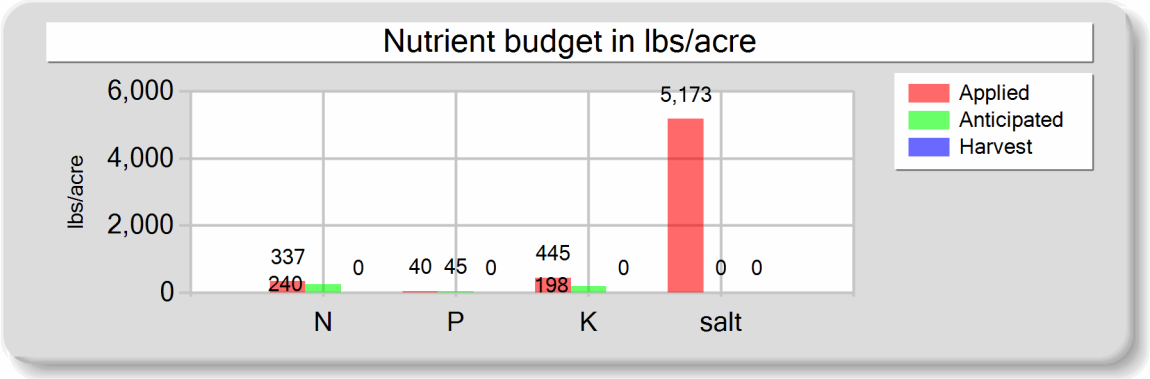
Fresh water applied	
4,560,000.00	*gallons*
167.93	*acre-inches*
4.20	*inches/acre*

Process wastewater applied	
576,000.00	*gallons*
21.21	*acre-inches*
0.53	*inches/acre*

Total harvests for the crop	
1	*harvests*

#3 - 06/15/2023: Corn, silage

Field name: #3 Crop: Corn, silage Plant date: 06/15/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	270.71	39.88	444.62	1,628.28
Fresh water	58.90	0.00	0.00	3,545.16
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	336.60	39.88	444.62	5,173.43
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	0.00	0.00	0.00	0.00
Nutrient balance	336.60	39.88	444.62	5,173.43
Applied to removed ratio	0.00	0.00	0.00	0.00

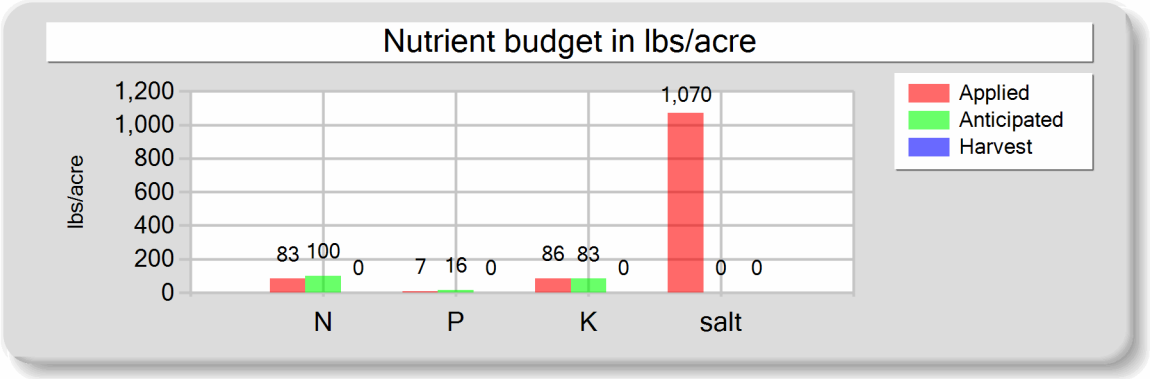
Fresh water applied	
27,408,000.00	*gallons*
1,009.34	*acre-inches*
25.23	*inches/acre*

Process wastewater applied	
3,096,000.00	*gallons*
114.02	*acre-inches*
2.85	*inches/acre*

Total harvests for the crop	
1	*harvests*

#4 - 11/05/2022: Oats, silage-soft dough

Field name: #4 Crop: Oats, silage-soft dough Plant date: 11/05/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	66.09	7.07	85.92	480.67
Fresh water	9.80	0.00	0.00	589.82
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	82.89	7.07	85.92	1,070.50
Anticipated crop nutrient removal	100.00	16.00	83.00	0.00
Actual crop nutrient removal	0.00	0.00	0.00	0.00
Nutrient balance	82.89	7.07	85.92	1,070.50
Applied to removed ratio	0.00	0.00	0.00	0.00

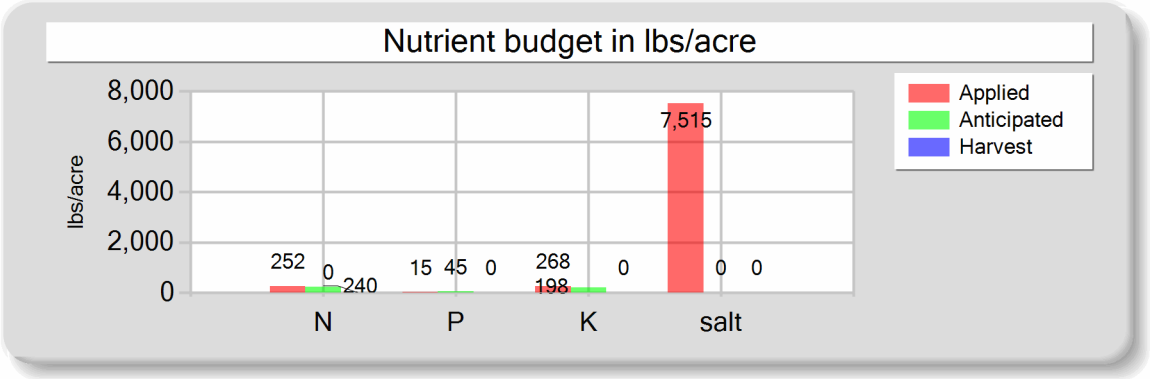
Fresh water applied	
1,140,000.00	*gallons*
41.98	*acre-inches*
4.20	*inches/acre*

Process wastewater applied	
144,000.00	*gallons*
5.30	*acre-inches*
0.53	*inches/acre*

Total harvests for the crop	
1	*harvests*

#4 - 06/15/2023: Corn, silage

Field name: #4 Crop: Corn, silage Plant date: 06/15/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	136.87	15.06	267.69	1,033.44
Fresh water	107.68	0.00	0.00	6,481.86
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	251.55	15.06	267.69	7,515.31
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	0.00	0.00	0.00	0.00
Nutrient balance	251.55	15.06	267.69	7,515.31
Applied to removed ratio	0.00	0.00	0.00	0.00

Fresh water applied	
12,528,000.00	*gallons*
461.36	*acre-inches*
46.14	*inches/acre*

Process wastewater applied	
456,000.00	*gallons*
16.79	*acre-inches*
1.68	*inches/acre*

Total harvests for the crop	
1	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

Manure 2305486-01

Sample and source description: Manure 2305486-01

Sample date: 05/30/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 32.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	7,240.00	2,290.00	15,100.00							43.70
DL	1.00	5.00	5.00							0.10

B. PROCESS WASTEWATER ANALYSES

Lagoon Water 2302053-01

Sample and source description: Lagoon Water 2302053-01

Sample date: 02/01/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	550.00	550.00			58.80	715.00								5,560.00	4,000
DL	0.60	0.60			0.10	10.00								1.00	1

Lagoon Water 2303398-01

Sample and source description: Lagoon Water 2303398-01

Sample date: 03/27/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	148.00	148.00			28.40	692.00								4,120.00	3,300
DL	0.60	5.00			0.10	10.00								1.00	1

Lagoon Water 2307235-01

Sample and source description: Lagoon Water 2307235-01

Sample date: 07/16/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	334.00	334.00			30.00	710.00								5,180.00	2,800
DL	0.60	0.60			0.10	10.00								1.00	1

Lagoon water 2310018-01

Sample and source description: Lagoon water 2310018-01

Sample date: 09/29/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	456.00	366.00			75.50	679.00								5,340.00	2,400
DL	0.60	0.60			0.10	10.00								1.00	1

C. FRESH WATER ANALYSES

Ag Well

2312554-03

Sample description: 2312554-03

Sample date: 12/29/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	10.30										852.00	620
DL	1.40										1.00	1

Domestic Well/ barn

Domestic Well/ barn

2312554-02

Sample description: 2312554-02

Sample date: 12/29/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	4.71										672.00	
DL	0.40										1.00	

Domestic Well/ house

2312554-01

Sample description: 2312554-01

Sample date: 12/23/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	4.53										643.00	
DL	0.40										1.00	

D. SOIL ANALYSES

No soil analyses entered.

E. PLANT TISSUE ANALYSES

#1 - 11/05/2022: Oats, silage-soft dough

#1 - 11/05/2022: Oats, silage-soft dough

2305485-01

Sample and source description: 2305485-01

Sample date: 05/30/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 54.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	11,100.00	1,600.00	10,500.00		10.30
DL	0.50	0.50	0.50		0.10

#2 - 11/05/2022: Oats, silage-soft dough

2305485-02

Sample and source description: 2305485-02

Sample date: 05/30/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 66.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	18,500.00	2,150.00	15,000.00		14.90
DL	0.50	0.50	0.50		0.10

#3 - 11/05/2022: Oats, silage-soft dough

2305485-03

Sample and source description: 2305485-03

Sample date: 05/30/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 63.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	15,200.00	2,050.00	14,700.00		12.90
DL	0.50	0.50	0.50		0.10

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

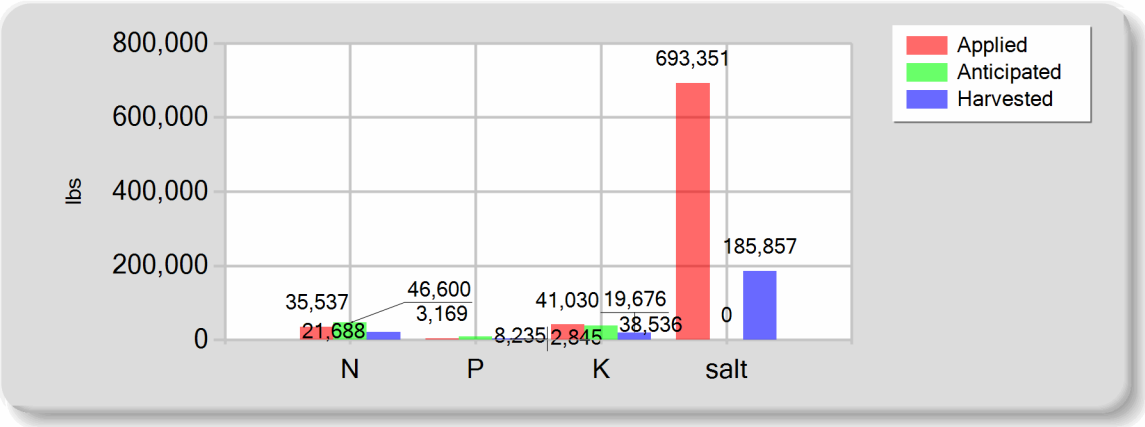
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

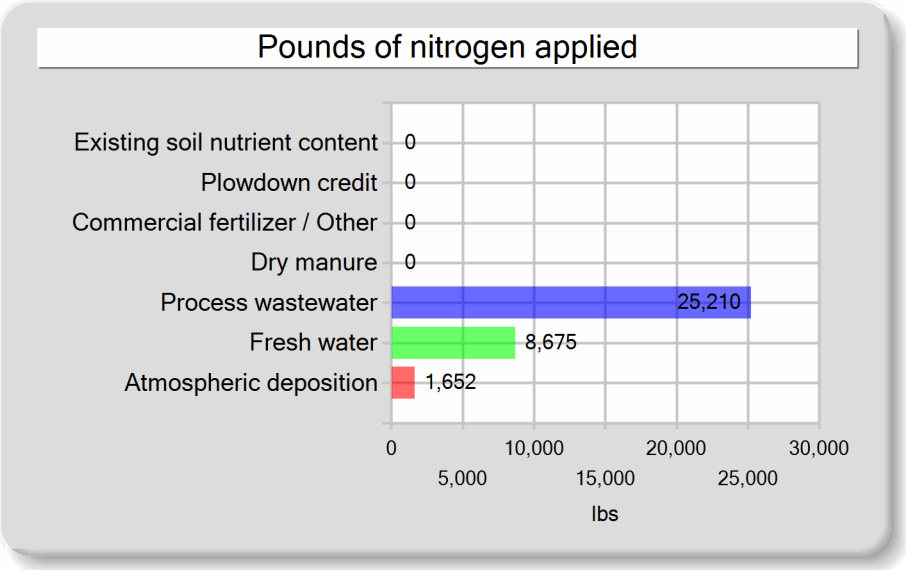
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

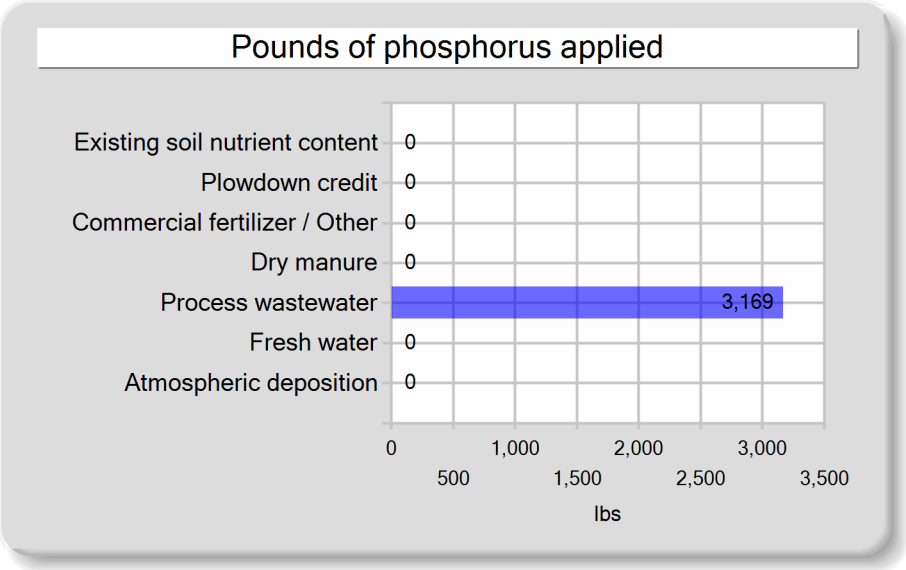
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	25,209.94	3,168.61	41,030.31	171,139.26
Fresh water	8,675.46	0.00	0.00	522,212.07
Atmospheric deposition	1,652.00	0.00	0.00	0.00
Total nutrients applied	35,537.40	3,168.61	41,030.31	693,351.33
Anticipated crop nutrient removal	46,600.00	8,234.80	38,536.40	0.00
Actual crop nutrient removal	21,687.77	2,844.79	19,675.82	185,856.77
Nutrient balance	13,849.64	323.82	21,354.49	507,494.57
Applied to removed ratio	1.64	1.11	2.09	3.73

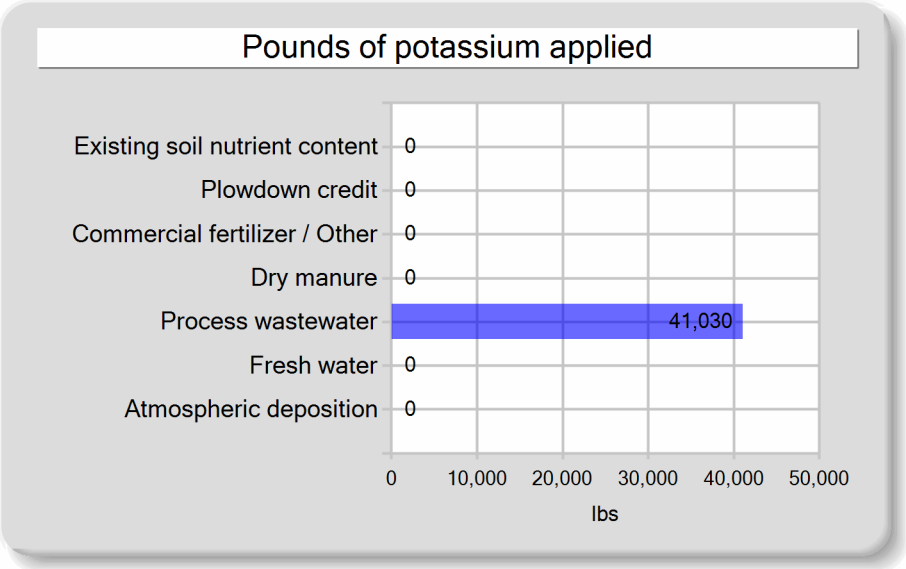
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

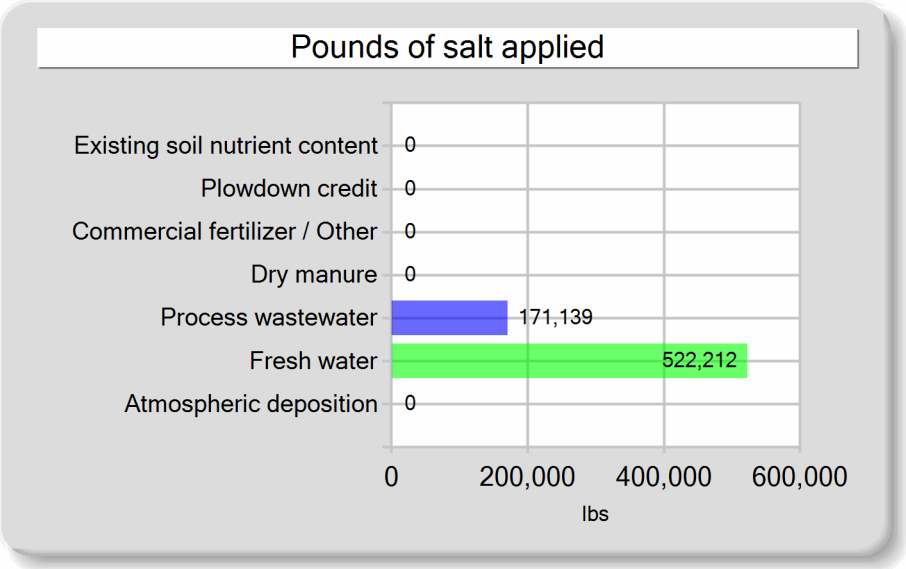


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	No
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

*Field #4 Oat Silage was not sampled in 2023, we will be sure to sample in 2024.

*Fields #1, #2 #3 & #4 Corn Silage was not sampled in 2023, we will be sure to sample in 2024.

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

Germano Soares	
SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
Germano L Soares	SAME AS OWNER
PRINT OR TYPE NAME	PRINT OR TYPE NAME
7/12/24	
DATE	DATE

ATTACHMENTS

A. REQUIRED ATTACHMENTS

The following lists the required documents that should be attached to the Annual Report when submitted .

Annual Dairy Facility Assessment	N/A
Provide an Annual Dairy Facility Assessment (an update to the Preliminary Dairy Facility Assessment in Attachment A) for each reporting period. On the PDFA Final page, click on the ADFA Report button to generate an ADFA report after updating information as needed .	

Manure/Process Wastewater Tracking Manifests	N/A
Provide copies of all manure/process wastewater tracking manifests for the reporting period, signed by both the owner/operator and the hauler.	

Corrective Actions Documents	NA
Provide records documenting any corrective actions taken to correct deficiencies noted as a result of the inspections required in the Monitoring Requirements of the General Order. Deficiencies not corrected in 30 days must be accompanied by an explanation of the factors preventing immediate correction.	

Groundwater Monitoring	N/A
Dischargers that monitor supply wells or subsurface (tile) drainage systems, or that have monitoring well systems must submit monitoring results as directed in the General Order, Groundwater Reporting Section starting on page MRP-13.	

Storm Water Monitoring	N/A
Dischargers that are required to monitor storm water more frequently than required in the General Order must submit monitoring results as directed in the General Order, Storm Water Reporting Section on page MRP-14.	